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Homework 6

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Contents

Problem 1: American Express Credit Card Applicants **1**

Recitation Exercise 1: Employment and the Great Recession (SW 11.1) **18**

Recitation Exercise 2: Workplace Smoking Bans (SW 11.2) **29**

Appendix: Tables and Figures **35**

Replication Files **48**

ECON UN3412 Spring 2026: Homework 6 Tyler Sotomayor

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Instructor: Seyhan Erden Due Date: 15 April 2026

Problem 1 [100 points]

This question deals with the dataset of American Express credit card applicants `AmEx.dta`. To be able to view the data set with STATA, you may need to write `set memory 32m` on the command window first. For each part, you will need to report results for the linear probability model, logit, and probit models. The best way to organize your results is in the form of a table. Report the log-likelihood and pseudo R^2 for the nonlinear models and the R^2 and adjusted R^2 for the linear model

cardhldr:	a binary indicator for whether or not the individual is approved as cardholder
income	the applicants income
age:	the applicants age in years
selfempl:	a binary indicator for whether or not the individual is self-employed
ownrent:	an indicator for whether or not an individual owns or rents their primary residence
acadmos:	the number of months and individual has resided at their current address
majdrg:	the number of major derogatory credit events in the last 12 months
mindrg:	the number of minor derogatory credit events in the last 12 months.

(a) [3 points] First, To get the scale of the coefficients right, `replace` *income* = *income/1000*. You only need a few of the variables from the data set. So, it is a good idea to keep those and not the rest. You can do this with the following command:

```
keep cardhldr income age selfempl ownrent acadmos majordrg minordrg
```

Now, what fraction of consumers in the sample are cardholders? What does this tell us in terms of external validity? (*hint*: use `tabulate` command)

SOLUTION:

The sample contains $n = 12,604$ applicants, of whom

$$\hat{p} = \frac{1}{n}\sum_i \texttt{cardhldr}_i = \boxed{0.7828}$$

or about 78.3% were approved as cardholders. The sample is therefore not a random cross-section of American consumers; every row is a completed AmEx application, which constrains what the coefficients we estimate can be read to say about the population. Consumers who never applied are absent, and those who did apply skew toward higher-income, more financially sophisticated households that perceived themselves as eligible for a premium charge card, so regression coefficients speak to the conditional approval process given applicant status rather than to the unconditional probability of being a cardholder in the U.S. population. Several of the covariates—especially `income` and the derogatory-credit counts—likely correlate with AmEx's unobserved internal credit score, which means that reading the coefficients causally ("giving applicant i a higher `income` would raise approval by $\hat{\beta}$") requires assumptions much stronger than those licensed by the sampling design (see Stock and Watson, 2020, §11.3). The resulting 78% approval rate is high precisely because applicants have already self-selected on observable eligibility before filing. ■

(b) [3 points] Consider predicting whether or not an individual is approved as a cardholder

cardhldr as a function of *income*, *age*, *selfempl*, *ownrent*, *acadmos*. Write commands for LPM, logit and probit.

SOLUTION:

Let $Y_i = \texttt{cardhldr}_i \in \{0,1\}$ and $X_i = (\texttt{inc}_i, \texttt{age}_i, \texttt{selfempl}_i, \texttt{ownrent}_i, \texttt{acadmos}_i)$, where $\texttt{inc} \equiv \texttt{income}/1000$. The three population models are

$$\begin{aligned}
\text{LPM:} \quad \Pr(Y_i = 1 \mid X_i) &= X_i'\beta, \\
\text{Probit:} \quad \Pr(Y_i = 1 \mid X_i) &= \Phi(X_i'\beta), \\
\text{Logit:} \quad \Pr(Y_i = 1 \mid X_i) &= \Lambda(X_i'\beta) = \frac{e^{X_i'\beta}}{1 + e^{X_i'\beta}},
\end{aligned}$$

with Φ the standard-normal CDF and Λ the logistic CDF. The corresponding Stata commands (using Huber–White robust standard errors throughout) are

```
* LPM (OLS with HC1 SEs)
reg    cardhldr inc age selfempl ownrent acadmos, robust
* Logit (MLE with robust sandwich SE)
logit  cardhldr inc age selfempl ownrent acadmos, vce(robust)
* Probit (MLE with robust sandwich SE)
probit cardhldr inc age selfempl ownrent acadmos, vce(robust)
```

Columns (1)–(3) of Table 1 in the Appendix report the estimates. All three models deliver the same qualitative story: approval probability rises in income and home ownership and falls with self-employment; age and address tenure are economically small. Because the reduced specification omits derogatory credit events, these coefficients are informative but likely suffer omitted-variable bias (corrected in Parts (f)–(g)). ■

(c) [8 points] Compute the marginal effect (how does $\Pr(Y = 1)$ change) for someone who is self-employed and for whether or not they rent or own their home. Evaluate this for an individual who has median values for their X variables (covariates).

	LPM	Logit	Probit
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

SOLUTION:

At the sample medians $(\texttt{inc}, \texttt{age}, \texttt{acadmos}) = (26.16,\ 31.50,\ 30.00)$, with the *other* dummy held at 0, the discrete change in predicted probability from switching `selfempl` or `ownrent` from 0 to 1 is

	LPM	Probit	Logit
$\Delta selfempl$	-0.1020	-0.1209	-0.1222
$\Delta ownrent$	$+0.0501$	$+0.0511$	$+0.0491$
$\%\Delta selfempl$	-13.55%	-15.99%	-16.13%
$\%\Delta ownrent$	$+6.66\%$	$+6.75\%$	$+6.48\%$

For the LPM the marginal effect of each dummy is identically its regression coefficient (no interactions with the remaining covariates); for probit and logit it depends on the baseline X through the index. Economically, a *self-employed* applicant at the median of income and age sees the approval probability fall by roughly 10–12 percentage points (around 14–16% in relative terms); *home own-*

ership raises it by about 5 percentage points ($\approx 7\%$). The three models agree to within ~ 2 pp, a textbook illustration of Angrist and Pischke (2009)'s observation that LPM and nonlinear latent-variable models produce nearly identical average partial effects in the interior of $[0, 1]$. ■

(d) [15 points] Repeat the previous exercise for someone whose income is at the 20% and 80% of the consumers in the sample.

	LPM	Logit	Probit
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

	LPM	Logit	Probit
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

SOLUTION:

The empirical 20th and 80th percentiles of `inc` are \$19.00k and \$38.80k, respectively. Replacing the income slot while keeping age and address tenure at their medians yields the marginal-effect profile shown in Table 2 (Appendix). The headline numbers are

	$\Delta\hat{P}$ (absolute)			$\%\Delta\hat{P}$ (relative)		
Income	LPM	Probit	Logit	LPM	Probit	Logit
Δ*selfempl*						
p20 ($19k)	−0.1020	−0.1274	−0.1317	−14.07%	−17.71%	−18.37%
p50 ($26k)	−0.1020	−0.1209	−0.1222	−13.55%	−15.99%	−16.13%
p80 ($39k)	−0.1020	−0.1068	−0.1025	−12.72%	−13.11%	−12.49%
Δ*ownrent*						
p20 ($19k)	+0.0501	+0.0555	+0.0549	+6.91%	+7.71%	+7.67%
p50 ($26k)	+0.0501	+0.0511	+0.0491	+6.66%	+6.75%	+6.48%
p80 ($39k)	+0.0501	+0.0428	+0.0388	+6.25%	+5.25%	+4.73%

The absolute LPM marginal effect is identical across income levels, as it must be: linearity implies $\Delta\hat{P} = \hat{\beta}$ regardless of the remaining covariates. For probit and logit the discrete changes shrink in magnitude as income rises: at the 20th percentile the conditional approval probability starts near 0.72, so the latent index sits squarely where $\phi(\cdot)$ and $\Lambda'(\cdot)$ are largest, while at the 80th percentile P_0 has risen to roughly 0.82, pushing the index into the right-hand tail where the link-function slope is smaller. That is precisely the nonlinearity the LPM cannot capture. The qualitative conclusion, however, survives across all three: self-employment is uniformly bad news (12–18% lower approval), home ownership uniformly good news (5–8% higher), and only the magnitude of each dummy's effect depends on income for the nonlinear models. ■

(e) [5 points] Express all three cases (20%, 50%, 80%) in terms of the probability ratio or log-odds. How big is the impact? Is this a lot or a little?

SOLUTION:

Re-expressing each marginal effect as a change in log-odds, $\Delta \log\left[\hat{P}/(1-\hat{P})\right]$, yields Table 3 in the Appendix. The headline values are

Change	Income	LPM	Probit	Logit
Δ*selfempl*	p20	-0.4668	-0.5691	-0.5847
	p50	-0.4908	-0.5771	-0.5847
	p80	-0.5513	-0.5959	-0.5847
Δ*ownrent*	p20	$+0.2677$	$+0.2944$	$+0.2894$
	p50	$+0.2907$	$+0.3006$	$+0.2894$
	p80	$+0.3521$	$+0.3140$	$+0.2894$

(Values are re-computed from the P_0, P_1 scalars stored during parts (c)–(d) of `ps6_amex.do`; see Table 3.) For the logit these quantities equal the dummy's regression coefficient up to rounding—-0.5847 for `selfempl` and $+0.2894$ for `ownrent`—and are invariant to the income slice, which is the columnar pattern the table shows. The LPM and probit values, by contrast, shift with the baseline income level because their implied log-odds change depends on where P_0 sits inside $(0, 1)$. The equivalent odds-ratios are

$$\boxed{\mathrm{OR}(\texttt{selfempl}) \;=\; e^{-0.5847} \approx 0.557, \quad \mathrm{OR}(\texttt{ownrent}) \;=\; e^{+0.2894} \approx 1.336}.$$

Is this a lot or a little? A lot. An odds ratio of 0.557 means self-employment cuts the odds of approval almost in half; 1.336 means home ownership raises the odds by a third. The economic magnitude is substantial but not extreme: both fall well inside the range where nonlinear and linear models agree on average

partial effects (Stock and Watson, 2020, §11.2). In a real underwriting context, a shift of that size would materially change application volume and risk-weighted revenue, so the impact is "a lot" in the practical sense of warranting underwriting attention even if "a little" in the statistical sense of being comfortably identified in moderate samples. ■

REFLECTIONS: The relative smallness of the LPM-to-logit-to-probit spread is striking: across six scenarios the three models' absolute $\Delta\hat{P}$ disagree by at most $\approx$ 3 percentage points. This is not a coincidence but a consequence of the approval rate sitting comfortably in the interior ($\approx$ 70–85%) at the covariate combinations we evaluate. Horrace and Oaxaca (2006) characterize exactly when the LPM breaks down: when $P(Y{=}1 \mid X)$ approaches the boundary, LPM predictions exit $[0, 1]$ and the implied marginal effects lose meaning.

(f) [4 points] Now consider a specification that includes major and minor derogatory credit events in the last year (*majdrg*,*mindrg*). Write commands for LPM, logit and probit.

SOLUTION:

The variable names in `AmEx.dta` are actually `majordrg` and `minordrg`. Appending both to the right-hand side gives

```
reg    cardhldr inc age selfempl ownrent acadmos majordrg minordrg, robust
logit  cardhldr inc age selfempl ownrent acadmos majordrg minordrg, vce(robust)
probit cardhldr inc age selfempl ownrent acadmos majordrg minordrg, vce(robust)
```

Columns (4)–(6) of Table 1 show these estimates. Adding the two derogatory-event counts fundamentally changes the story: the LPM's R^2 jumps from 0.0269 to 0.2118, and the logit/probit pseudo-R^2 both jump from $\approx$ 0.029 to $\approx$ 0.23—an

eight-fold improvement that dwarfs what demographic controls alone delivered in (b). ■

(g) [18 points] Report the full table of specifications (there should be 6) with all relevant statistics. Mark 1%, 5% and 10% significance levels using the key at the bottom of the table.

VARIABLES	(1) *LPM*	(2) *Logit*	(3) *Probit*	(4) *LPM*	(5) *Logit*	(6) *Probit*
income	( )	( )	( )	( )	( )	( )
age	( )	( )	( )	( )	( )	( )
selfemp	( )	( )	( )	( )	( )	( )
ownrent	( )	( )	( )	( )	( )	( )
acadmos	( )	( )	( )	( )	( )	( )
majordrg	–	–	–	( )	( )	( )
minordgr	–	–	–	( )	( )	( )
Constant	( )	( )	( )	( )	( )	( )
Observations						
R^2		–	–		–	–
Pseudo R^2	–			–		

Robust standard errors in parentheses

$^{**}\, p < 0.01,\ ^{*}\, p < 0.05,\ ^{+}\, p < 0.1$

Solution:

The six-column table is in the Appendix (Table 1), and several features of that specification deserve attention. The derogatory-event controls dominate the expanded specification. In the LPM, each additional major derogatory event cuts the approval probability by 11.7 pp ($p < 0.01$); each minor event by 3.2 pp; and the logit and probit estimates imply very similar average marginal effects,

which part (j) takes up directly. The income effect survives intact—the coefficient on `inc` (per \$1,000) moves only from 0.00390 to 0.00393 in the LPM—confirming that income and credit-history are largely orthogonal as sources of underwriting information. The two dummies are similarly stable: `selfempl` stays at roughly -10 pp and `ownrent` at roughly $+4.5$ pp in the LPM. The change that does occur is in the age coefficient, which collapses from -0.00130 ($p = 0.003$) in the reduced model to statistically insignificant once derogatory events are present ($p = 0.55$ for the LPM, $p \approx 0.70$ for probit and logit), indicating that the "age effect" in parts (b)–(c) was largely proxying for accumulated derogatory marks on older applicants' records. The one coefficient that gains significance is `acadmos`: address tenure reaches the 1% level in the full model, picking up a modest but real stability-of-residence signal once credit history is held fixed. ■

(h) [6 points] Which model fits the data the best? Which variables have the most explanatory power? Do major derogatory credit events matter? Do you prefer models with or without derogatory events?

Solution:

Among the three full-specification models, the logit narrowly wins on every fit criterion: a log-likelihood of -5066.37 versus -5080.81 for the probit, and a pseudo-R^2 of 0.2318 versus 0.2296. The LPM, whose $R^2 = 0.2118$ and RMSE 0.366 are not directly comparable, trails but agrees qualitatively. The probit and logit full-model fits are essentially tied, which is the expected pattern when the link functions are close to linear on the bulk of the sample; what all three models agree on is that introducing `majordrg` and `minordrg` multiplies the fraction of variance in `cardhldr` that can be explained by roughly eight.

The variable carrying the most explanatory power is unambiguously `majordrg`. Its standardized coefficient in the LPM is $-0.117 \times 1.44 \approx -0.17$, roughly twice that of `selfempl` or `ownrent`; its t-statistic is the most extreme in the table ($|t| \approx 27$ in the LPM, $|z| \approx 33$ in probit and logit); and its coefficient is also by far the largest in the full-model column, dwarfing the -0.0317 coefficient on `minordrg`. Major derogatory events unambiguously matter, reducing approval probability by roughly 11–17 pp per event depending on the evaluation point, as part (j) quantifies at median covariates.

I prefer the model that includes the derogatory-event counts. The reduced specification suffers from severe omitted-variable bias—the `selfempl` and `age` coefficients shrink (`age` collapses to zero), and the R^2 is an order of magnitude smaller—and, substantively, any credible account of credit-card approval has to include prior credit behavior. The data back up that substantive prior overwhelmingly. ■

(i) [6 points] Construct a 95% confidence interval for the effect of an additional major derogatory credit event has on being an American Express cardholder under the linear probability model.

SOLUTION:

From Table 1 column (4), $\hat{\beta}^{\text{LPM}}_{\texttt{majordrg}} = -0.1175$ with heteroskedasticity-robust standard error 0.0043. The 95% normal-approximation interval is

$$\hat{\beta} \pm 1.96 \cdot \text{SE}(\hat{\beta}) \;=\; -0.1175 \pm 1.96 \cdot 0.0043 \;=\; \boxed{[-0.1259,\ -0.1090]}.$$

Each additional major derogatory event is estimated to lower the probability of

approval by 10.9–12.6 percentage points, with the effect precisely pinned down because `majordrg` varies substantially in the sample (standard deviation 1.44, max 22). The interval excludes zero by a very wide margin: the t-statistic on `majordrg` is -27.24. ■

(j) [6 points] Repeat for the logit and probit models in the case where the x values take on the median values in the sample.

SOLUTION:

For logit and probit the coefficient itself is not a marginal effect. Using `margins, dydx(majordrg)` at the sample medians $(\texttt{inc}, \texttt{age}, \texttt{acadmos}, \texttt{majordrg}, \texttt{minordrg}) = (26.16, 31.50, 30.00, 0, 0)$ and the remaining dummies set to 0:

Model	$\widehat{\partial P}/\partial$majordrg	SE	95% CI
LPM	-0.1175	0.0043	$[-0.1259,\ -0.1090]$
Probit	-0.1715	0.0054	$[-0.1821,\ -0.1609]$
Logit	-0.1618	0.0054	$[-0.1724,\ -0.1512]$

(Standard errors are delta-method robust sandwich SEs reported by Stata's `margins` with `vce(robust)`.) At median covariates, probit and logit both estimate a 16–17 percentage-point decline per major derogatory event, roughly 50% larger in absolute value than the LPM coefficient. The difference is unsurprising: at $\hat{P}_0 = 0.76$, the index is already well off the 0.5 midpoint, so the local slope of Φ or Λ exceeds the LPM's constant slope near this baseline. All three intervals are tight and exclude zero. ■

Reflections: The probit/logit marginal effect at the *median* is not the only reasonable summary. Stata's `margins, dydx(*)` with no `at()` option delivers the average partial effect (APE)—the mean of the individual-specific marginal effects across the estimation sample—which often differs noticeably from the effect at the median in highly skewed data. Computed on this sample, the APE of `majordrg` is -0.1584 for the probit and -0.1536 for the logit (delta-method SEs 0.0041 and 0.0040), roughly 30–35% larger in magnitude than the LPM coefficient of -0.1175. The gap is precisely the tail-region story diagnosed in part (n): `majordrg` pushes many applicants' approval probabilities close to zero, and averaging local slopes over the full covariate support picks up those steeper tail regions, whereas the LPM coefficient is closer to the linear average.

(k) [8 points] Compute the fitted values for the linear probability model, and plot a histogram for the fitted values of the y variable. Do the same for logit and probit models.

Solution:

After estimating each full-controls model, `predict` produces fitted values: `xb` for the LPM (linear index) and `pr` for the probit and logit (predicted probabilities on $[0, 1]$). Figure 1 in the Appendix plots the three histograms side-by-side.

The LPM panel visibly spills off both boundaries: $\min(\hat{P}^{\text{LPM}}) = -1.8271$ and $\max(\hat{P}^{\text{LPM}}) = 1.1861$, well outside the legitimate $[0, 1]$ range. Probit and logit fitted probabilities are bounded by construction and concentrate tightly around $P \in [0.7, 1.0]$, with a small second lump near 0.1–0.3 corresponding to applicants with several derogatory events. The two nonlinear histograms are almost visu-

ally indistinguishable—again consistent with the interior-of-probability agreement between probit and logit. ■

(l) [6 points] What should be the domain of predicted values from the LPM? What fraction of values lies outside this domain? How do we interpret them?

SOLUTION:

Because Y is a probability, the legitimate domain of any fitted value is $[0, 1]$. The LPM does not respect that constraint: the fitted index $\hat{P}_i = X_i'\hat{\beta}$ is a linear function of the covariates and will take arbitrarily extreme values when some regressor is far from its mean. In this sample 0.0117 of observations have $\hat{P}^{\text{LPM}} < 0$ and 0.0288 have $\hat{P}^{\text{LPM}} > 1$, with fitted values spanning -1.8271 to 1.1861. In total,

$$\boxed{\text{fraction of LPM fitted values outside } [0,1] \;=\; 0.0405}.$$

These out-of-range predictions cannot be interpreted as probabilities. A value of -1.8 simply means that the linear predictor is highly negative, which is itself a signal that the applicant has so many derogatory events (or so little income) that the true approval probability is essentially zero. Standard practical responses are to truncate the prediction at 0 or 1 (simple but biased in finite samples), to switch to a nonlinear model whose link function guarantees $\hat{P} \in (0, 1)$ (the probit or logit approach), or to fit a fractional logit (see Angrist and Pischke, 2009) which marries the LPM's computational simplicity to the logistic CDF's bounded range. The fact that roughly 4% of this sample produces nonsensical LPM predictions is one of the standard textbook arguments for probit or logit

whenever the outcome is highly unbalanced or the regressors are dispersed. ■

(m) [6 points] From the results you find on Table 1, what can you say about probit and logit results? How about LPM results?

SOLUTION:

Despite using different link functions, probit and logit deliver almost identical conclusions: the signs of every coefficient agree, every regressor gets the same significance verdict (both models flag `minordrg` as marginally significant at the 10% level and every other regressor at the 1% level), the implied marginal effects at the median differ by roughly one percentage point for `majordrg` (-0.1715 probit vs. -0.1618 logit, a gap of about 0.01) and by about 0.002 for the dummies, and the log-likelihoods and pseudo-R^2 are essentially tied at -5066.4 versus -5080.8 and 0.232 versus 0.230. The logit coefficients are roughly $\sqrt{\pi^2/3} \approx 1.81$ times the probit coefficients—the familiar rescaling owing to different error variances in the two latent-variable formulations—but the implied probabilities are nearly identical once that scaling is absorbed by the link function.

The LPM gets the sign and significance pattern exactly right, and its coefficients are easy to read directly as marginal effects. Its two shortcomings are documented elsewhere in the solution set: roughly 4% of its fitted values escape $[0, 1]$ (part (l)), and its constant slope cannot reproduce the dependence of the nonlinear models' marginal effects on the baseline X (part (d)). For most coefficients in the specification the LPM estimate is close to the probit and logit APEs, but `majordrg` is a visible exception: its APE is -0.1584 (probit) and -0.1536 (logit) versus an LPM coefficient of -0.1175, a gap of roughly 3–4 percentage

points that the Amemiya rule check in part (n) attributes to `majordrg` pushing fitted probabilities into the tails. For accurate prediction of individual probabilities, especially for applicants with high `majordrg` or low `inc`, the LPM is inadequate and probit or logit should be preferred. ■

(n) [6 points] A well-known result by Amemiya (1981) is that over the range of probabilities from 30% to 70% the following approximations should hold: $\beta_{\text{LPM}} \approx 0.4\,\beta_{\text{probit}}$ and $\beta_{\text{LPM}} \approx 0.25\,\beta_{\text{logit}}$. Take your results for logit and probit, how does this result look for the $\beta_{majordrg}$?

SOLUTION:

The relevant coefficients from Table 1 are

$$\hat{\beta}^{\text{LPM}}_{\texttt{majordrg}} = -0.1175, \qquad \hat{\beta}^{\text{probit}}_{\texttt{majordrg}} = -0.7059, \qquad \hat{\beta}^{\text{logit}}_{\texttt{majordrg}} = -1.2344.$$

The Amemiya-implied approximations are

$$0.40 \cdot \hat{\beta}^{\text{probit}} = -0.2824, \qquad 0.25 \cdot \hat{\beta}^{\text{logit}} = -0.3086,$$

both substantially larger in absolute value than the LPM coefficient. The actual ratios are

$$\boxed{\frac{\hat{\beta}^{\text{LPM}}}{\hat{\beta}^{\text{probit}}} = 0.1664 \ \ (\text{rule } 0.40), \qquad \frac{\hat{\beta}^{\text{LPM}}}{\hat{\beta}^{\text{logit}}} = 0.0951 \ \ (\text{rule } 0.25)}.$$

The rule fails badly here because the Amemiya (1981) approximation is built on the link-function derivative in the range $P \in [0.3, 0.7]$—the interior where

$\phi(z)$ and $\Lambda'(z)$ are largest—and `majordrg` has a long right tail (max = 22) that pushes many applicants' approval probabilities to essentially zero, far outside that band. The LPM's coefficient averages over the full covariate support, whereas the probit and logit coefficients capture the latent-index slope even in regions where the CDF is near-vertical; the two are not directly comparable when a regressor drives fitted probabilities into the extremes. The 0.4/0.25 rule is best read as a rough internal-consistency check rather than a law: it tends to break down precisely when a regressor generates large probability variation, which is exactly when we care most about the coefficient. As Angrist and Pischke (2009, Ch. 3) note, the LPM coefficient is better compared to the average partial effect from probit/logit than to the rescaled index coefficient—which is the quantity the Amemiya rule targets. ■

Reflections: The breakdown of the Amemiya rule for `majordrg` is instructive. Reapplying the same rule to the *dummy* variables (where the probability shifts stay inside the interior band) produces closer agreement: for `selfempl`, $0.4 \times (-0.445) = -0.178$ vs. LPM -0.100 ($\approx 56\%$ of the predicted value); for `ownrent`, $0.4 \times 0.191 = 0.0764$ vs. LPM 0.0456 ($\approx 60\%$). Still below the target, but far closer than the 17–10% we see for `majordrg`. The divergence is thus specifically a feature of regressors that push predicted probabilities into the tails.

Recitation Exercise 1 (SW Empirical Exercise 11.1, ungraded)

In April 2008, the unemployment rate in the United States stood at 5.0%. By April 2009, it had increased to 9.0%, and it had increased further, to 10.0% by October 2009. Were some groups of workers more likely to lose their jobs than others during the Great Recession? For example, where young workers more likely to lose their jobs than middle-aged workers? What about workers with a college degree versus those without a degree or women versus men. We will be working with `employment_08_09.dta`, which contains a random sample of 5,440 workers who were surveyed in April 2008 and reported that they were employed full time. The workers were surveyed one year later, in April 2009, and asked about their employment status (employed, unemployed, or out of the labor force).

(a) [8 points] What fraction of workers in the sample were employed in April 2009? Use your answer to compute a 95% confidence interval for the probability that a worker was employed in April 2009, conditional on being employed in April 2008.

SOLUTION:

The sample consists of $n = 5{,}412$ workers who were employed full time in April 2008. The binary variable `employed` takes the value 1 if the same worker was employed (in any capacity) in April 2009 and 0 otherwise. Its sample mean is therefore the sample analogue of the conditional probability of interest, $\Pr(\text{Empl}_{09} = 1 \mid \text{Empl}_{08} = 1)$:

$$\hat{p} \;=\; \frac{1}{n}\sum_{i=1}^{n} \text{employed}_i \;=\; 0.8755.$$

Because employed$_i$ is Bernoulli, the large-sample standard error of $\hat{p}$ is

$$\mathrm{SE}(\hat{p}) \;=\; \sqrt{\tfrac{\hat{p}(1-\hat{p})}{n}} \;=\; \sqrt{\tfrac{0.8755\times 0.1245}{5{,}412}} \;=\; 0.00449.$$

Applying the standard 95% normal approximation yields

$$\hat{p} \pm 1.96\,\mathrm{SE}(\hat{p}) \;=\; 0.8755 \pm 0.0088 \;=\; \boxed{[\,0.8667,\ 0.8843\,]}.$$

Roughly 86.7–88.4% of workers employed in April 2008 were still employed one year later. The complement of this interval, $[11.6\%, 13.3\%]$, gives the fraction who lost their job or left the labor force during the recessionary year. See Stock and Watson (2020), §11.1, for the underlying binary-outcome setup. ■

Reflections: The Bernoulli SE formula is a special case of the Wald standard error for a binary-response model with only an intercept: in the LPM of `employed` on a constant, the OLS coefficient equals $\hat{p}$ and the homoskedasticity-consistent SE is $\sqrt{\hat{p}(1-\hat{p})/n}$. Because the population variance of a Bernoulli equals $p(1-p)$, this SE is *not* obviously biased in small samples—it is the maximum-likelihood SE—and the Wilson or exact Clopper–Pearson interval would yield only a marginally different CI here given $n = 5{,}412$.

(b) [8 points] **(b.i & ii)** Regress *Employed* on *Age* and Age^2, using a linear probability model. Based on this regression, was age a statistically significant determinant of employment in April 2009? Is there evidence of a nonlinear effect of age on the probability of being employed?

SOLUTION:

Let E_i denote `employed`. The linear probability model is

$$E_i \;=\; \beta_0 + \beta_1 \,\text{age}_i + \beta_2 \,\text{age}_i^2 + u_i,$$

fit by OLS with heteroskedasticity-robust (HC1) standard errors; heteroskedasticity is inherent to the LPM because $\text{Var}(E_i \mid X_i) = \Pr(E_i = 1 \mid X_i)\,[1 - \Pr(E_i = 1 \mid X_i)]$ varies with X_i (Stock and Watson 2020, §11.1).

The estimated equation is

$$\boxed{\widehat{\text{employed}}_i = \underset{(0.0669)}{0.307} + \underset{(0.00328)}{0.0283}\,\text{age}_i \;-\; \underset{(0.0000388)}{0.000327}\,\text{age}_i^2},$$

with $n = 5{,}412$ and $R^2 = 0.0197$. Individually, $t_{\beta_1} = 8.62$ and $t_{\beta_2} = -8.42$, each with $p < 0.001$. Jointly, the robust F-statistic for $H_0 : \beta_1 = \beta_2 = 0$ is $F(2, 5409) = 37.50$ with $p < 0.001$.

Age is therefore a statistically significant determinant of April 2009 employment at any conventional level, and because $\hat{\beta}_2$ is itself significantly different from zero the relationship is nonlinear: the quadratic specification fits strictly better than the linear-only one. Setting $\partial \hat{E}_i / \partial \text{age} = 0$ locates the implied peak at $-\hat{\beta}_1/(2\hat{\beta}_2) = 0.0283/(2 \times 0.000327) \approx 43.3$ years, with retention rising from the teens to the early 40s and then declining into the 60s. ■

(c) [4 points] **(b.iii)** Using the same LPM on *Age* and Age^2, compute the predicted probability of employment for a 20-year-old worker, a 40-year-old worker, and a 60-year-old worker.

SOLUTION:

Plugging into the estimated equation (or equivalently using `margins, at(age=(20 40 60))`):

Age	20	40	60
$\widehat{\Pr}(\text{Empl}_{09} = 1)$	0.7423	0.9158	0.8279

These values match the inverted-U pattern identified in part (b.i&ii): retention is lowest at the ages that bracket the peak and highest near it. Figure 2 plots the entire predicted age profile over $[18, 63]$ for all three models. ■

(d) [6 points] **(c)** Repeat (b) using a probit regression.

SOLUTION:

The probit model writes

$$\Pr(E_i = 1 \mid X_i) = \Phi(\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{age}_i^2),$$

where Φ is the standard normal CDF. Maximum-likelihood estimation with the Huber-White sandwich variance gives

$$\boxed{\widehat{\Pr}(\text{Empl}_{09} = 1) = \Phi\left(\underset{(0.250)}{-1.258} + \underset{(0.0129)}{0.122}\,\text{age}_i - \underset{(0.000155)}{0.00141}\,\text{age}_i^2\right)}.$$

The pseudo-R^2 (McFadden) is 0.0232. Individually, $z_{\beta_1} = 9.44$ and $z_{\beta_2} = -9.09$; jointly $\chi^2(2) = 92.89$, $p < 0.001$. Age is highly significant and nonlinear. The

predicted probabilities are

$$\widehat{\Pr}(E = 1 \mid \text{age} = 20, 40, 60) = 0.7296,\ 0.9117,\ 0.8316.$$

■

(e) [6 points] **(d)** Repeat (b) using a logit regression.

SOLUTION:

The logit model replaces Φ with the logistic CDF $\Lambda(\cdot) = e^{\cdot}/(1 + e^{\cdot})$:

$$\boxed{\widehat{\Pr}(\text{Empl}_{09} = 1) = \Lambda\left(\underset{(0.446)}{-2.490} + \underset{(0.0235)}{0.225}\,\text{age}_i - \underset{(0.000285)}{0.00262}\,\text{age}_i^2\right)}.$$

Pseudo-$R^2 = 0.0235$; $z_{\beta_1} = 9.61$, $z_{\beta_2} = -9.21$; joint $\chi^2(2) = 97.85$, $p < 0.001$.

Predicted probabilities at ages $20, 40, 60$ are 0.7251, 0.9114, 0.8310.

The full regression output for parts (b)–(d) is in Table 4 in the Appendix. ■

(f) [6 points] **(e)** Are there important differences in your answers to (b)–(d)? Explain.

SOLUTION:

The three models deliver nearly identical predicted probabilities:

Age	LPM	Probit	Logit
20	0.7423	0.7296	0.7251
40	0.9158	0.9117	0.9114
60	0.8279	0.8316	0.8310

The maximum spread across models is about 1.7 percentage points (at age 20); at ages near the sample mean (40), the three models agree to within ≤ 0.5 percentage points. Figure 2 confirms that the three predicted age profiles are virtually indistinguishable over the interior of the age distribution.

Two reasons account for this agreement. First, the sample is dominated by values of $\Pr(Y = 1)$ in the 0.7–0.95 range, well away from the boundaries where the LPM's linearity is most damaging. Second, over this interior range the probit and logit link functions are themselves nearly linear after appropriate rescaling—Angrist and Pischke (2009) note that $\hat{\beta}_{\text{LPM}}$ is generally well-approximated by the average partial effect from probit or logit, which is precisely what this figure shows. LPM does, however, slightly overstate $\widehat{\Pr}(E = 1)$ at age 20 relative to the nonlinear models, which is the familiar symptom of the LPM pulling probabilities above 1 and below 0 at extreme covariates. ■

Reflections: A useful rule of thumb from Cameron and Trivedi (2005) and Amemiya (1981) is that in the interior of the probability distribution the LPM slope is approximately $0.4\beta_{\text{probit}}$ and $0.25\beta_{\text{logit}}$. Plugging in the *age* coefficients, $0.4 \times 0.122 = 0.049$ and $0.25 \times 0.225 = 0.056$, versus the LPM's 0.0283—larger than the rule suggests, but not wildly so given the quadratic term. The rule is most reliable when the event is common but not rare; here it is in the 88% range, which is near the upper edge of "interior".

(g) [10 points] **(f.i & ii)** The data set includes variables measuring the workers' educational attainment, sex, race, marital status, region of the country, and weekly earnings in April 2008. Construct a table like Table 11.2 to investigate whether the conclusions on the effect of age on employment from (b)–(d) are affected by omitted variable bias.

Use the regressions in your table to discuss the characteristics of workers who were hurt most by the Great Recession.

SOLUTION:

Table 5 reports LPM, probit, and logit estimates for the same sample with the expanded controls. The LPM age coefficient falls only slightly when the additional covariates are added, from $\hat{\beta}_{\text{age}} = 0.0283$ to 0.0253, and $\hat{\beta}_{\text{age}^2}$ moves correspondingly from -0.000327 to -0.000293. The qualitative conclusion—that retention is an inverted-U in age, peaking in the early 40s—is therefore robust to including the full set of demographic and earnings controls, which suggests that the age profile identified in (b)–(d) was not merely reflecting composition effects along education, race, or earnings. Analogous stability holds for probit and logit.

Reading the LPM column, whose coefficients translate directly into probability changes, education emerges as the single strongest predictor of recession-era retention: relative to advanced-degree holders, high-school dropouts had a retention probability 8.8 pp lower ($p < 0.01$), with the other education categories also pointing downward if not always significantly so. Union members also had a substantially lower retention probability (-5.77 pp, $p < 0.01$), an effect that is most naturally read as sectoral exposure—union-intensive sectors like manufacturing, construction, and autos were hardest hit in 2008–09, and the union dummy captures that exposure even after demographics are held fixed. Earnings work in the expected direction, with each additional \$100/week in April 2008 earnings raising retention by 0.35 pp ($p < 0.01$), so workers at the bottom of the earnings distribution were correspondingly more vulnerable. The quadratic age profile

(Figure 2) traces out the familiar pattern: retention is lowest for 18–20-year-olds and falls again past age 55. Geographically, workers in central states were *more* likely to retain employment (+4.1 pp, $p < 0.01$), consistent with the 2008–09 recession being concentrated in the coastal finance and housing hubs rather than in the interior. Sex, race, and marital status have quantitatively small and mostly insignificant coefficients once the other factors are held fixed, which is one piece of evidence that the Great Recession did not systematically discriminate along those lines conditional on industry and occupation composition (captured here through union status and earnings). ■

(h) [6 points] **(g)** The results in (a)–(f) were based on the probability of employment. Workers who are not employed can either be (i) unemployed or (ii) out the labor force. Do the conclusions you reached in (a)–(f) also hold for workers who became unemployed? (Hint: Use the binary variable *Unemployed* instead of *Employed*.)

SOLUTION:

Switching the dependent variable to `unemployed`, which equals 1 for the 239 workers who were actively unemployed in April 2009 and 0 for both the employed and the out-of-labor-force groups, flips the sign pattern in the expected ways; Table 6 reports the results. The age coefficient reverses from positive to negative ($\hat{\beta}_{\text{age}} = -0.00506$ in the LPM), and $\texttt{age}^2$ reverses from negative to positive, so the age profile becomes a U: young workers are most likely to *become unemployed*, middle-aged workers least so, with a modest uptick toward the 60s. That mirror image of the employment profile is exactly what a labor-market flow interpretation would predict. Education delivers a particularly sharp gradient on this margin: high-school dropouts are a remarkable 6.99 pp *more* likely

to be unemployed in April 2009 than advanced-degree holders ($p < 0.01$), and high-school graduates are 3.39 pp more likely—the monotone education gradient from part (f) becomes even steeper for unemployment than for retention, which confirms education as the first-order determinant of recession vulnerability.

The demographic margins do not translate as cleanly. Women are in fact *less* likely to be classified as unemployed (-2.02 pp, $p < 0.01$), but that is largely a labor-force attachment story rather than a retention story: women who lost jobs more often exited the labor force entirely rather than actively searching, and the "unemployed" dummy only captures the latter. Married workers are similarly less likely to be actively unemployed, conditional on having lost their job. The union coefficient is statistically indistinguishable from zero in the unemployment equation even though it was -5.77 pp in the employment equation—consistent with unemployed union members tending to drop out of the labor force (or transition to early retirement) rather than be caught in the April 2009 active-unemployment snapshot. The earnings coefficient also flips to effectively zero: $\hat{\beta}_{\text{earnwk}} = -0.000378$ with SE 0.000645 in the LPM (t-statistic under unity), so earnings—which was a strong predictor of *retention* in part (f)—does not predict *active unemployment* with any precision in this sample. This is consistent with the same labor-force-exit mechanism: lower-earning workers who lose their jobs tend to leave the labor force rather than remain officially unemployed. The education and age conclusions from parts (a)–(f) therefore survive; the sex, union, and earnings patterns depend crucially on which margin we study—separation, active unemployment, or labor-force exit. ■

(i) [8 points] **(h)** These results have covered employment transitions during the Great Recession, but what about transitions during normal times? On the text website, you

will find the data file *Employment_06_07*, which measures the same variables but for the years 2006–2007. Analyze these data and comment on the differences in employment transitions during recessions and normal times.

SOLUTION:

In `employment_06_07.dta` (n = 5,220), the sample proportion of workers still employed one year later is $\hat{p}_{06} = 0.8977$ with 95% CI $[0.8895, 0.9059]$, or approximately 2.2 percentage points higher than the recession-era retention rate of 0.8755. Unemployment, in turn, is only 1.65% in the normal sample versus 4.42% in the recession sample—a nearly three-fold increase.

Estimating the same full-controls specification on the 2006–2007 sample (Table 7, visualized in Figure 3) delivers three instructive contrasts with the recession period. The age profile is flatter in normal times: predicted retention ranges from about 0.73 at age 18 to a peak of 0.93 near age 45 and falls to 0.83 at age 63, a total swing of roughly 20 pp, whereas the 2008–2009 profile starts lower (0.68 at age 18), peaks lower (around 0.915 near age 43), and falls further (to roughly 0.79 at age 63) for a swing of about 24 pp. The recession therefore both lowered the entire retention profile and steepened it, hurting the young and the old disproportionately. The education gradient is also smaller in normal times: high-school dropouts' retention penalty is −5.07 pp before the downturn versus −8.82 pp during it, a 75% amplification that makes education a stronger "moat" against layoff in a recession. The female effect operates in the opposite direction: women carry a larger retention penalty in normal times (−3.35 pp, $p < 0.01$) than in the recession (−0.66 pp, not significant), consistent with the "he-cession" observation that the 2008–09 downturn disproportionately struck male-

dominated industries—construction, manufacturing, finance—and temporarily narrowed the male–female separation gap.

The earnings coefficient is almost identical across the two periods (0.00357 vs. 0.00348 per \$100/week), suggesting that the earnings-to-retention pipeline is a stable feature of the labor market rather than a recession-specific phenomenon. Overall the recession did not change *who* faced higher job-loss risk; it amplified the gap between the vulnerable and the protected, and the gap widened most along the education axis. ■

Reflections: An important caveat is that the two samples are cross-sections of different workers, so differences in estimated coefficients reflect both genuine economic changes (recession vs. expansion) and sampling variation. A more rigorous differences-in-differences design would interact each regressor with a recession indicator and pool the two samples—this is a natural extension but is beyond the scope of SW 11.1. The qualitative conclusions, however, are large enough to be unlikely artifacts of sampling noise.

Recitation Exercise 2 (SW Empirical Exercise 11.2, ungraded)

Workers used to be able to smoke inside office buildings (now some students vape inside classrooms). Smoking bans were introduced in several office areas during the 1990s. Supporters of these bans argued that in addition to eliminating the externality of secondhand smoke, they would encourage smokers to quit by reducing their opportunities to smoke. We will use `smoking.dta`, which has data on 10,000 U.S. indoor workers from 1991 to 1993. The data set contains information on whether individuals were subject to a workplace smoking ban, whether the individuals themselves smoked, and other characteristics.

(a) [8 points] Estimate the probability of smoking for (i) all workers, (ii) workers affected by workplace smoking bans, and (iii) workers not affected by workplace smoking bans.

SOLUTION:

The sample analogues of these three conditional probabilities are the corresponding sample means of the binary variable `smoker`:

Group	$\hat{p}$	SE($\hat{p}$)	n
All workers	0.2423	0.0043	10,000
Workers with a workplace ban (`smkban` = 1)	0.2120	0.0052	6,098
Workers without a workplace ban (`smkban` = 0)	0.2896	0.0073	3,902

The standard errors are the Bernoulli formula applied within each cell. The raw difference in smoking rates is $\hat{p}_{\text{no ban}} - \hat{p}_{\text{ban}} = 0.2896 - 0.2120 = 0.0776$, or about a 7.8-percentage-point higher smoking rate among workers not covered by a ban. The key causal question—whether this difference reflects the ban's

effect or pre-existing selection of smokers into non-ban workplaces—is addressed in part (c). ■

(b) [8 points] What is the difference in the probability of smoking between workers affected by a workplace smoking ban and workers not affected by a workplace smoking ban? Use a linear probability model to determine whether this difference is statistically significant.

SOLUTION:

Regressing `smoker` on `smkban` alone yields

$$\boxed{\widehat{\Pr}(\text{smoker} = 1) = \underset{(0.00726)}{0.290} - \underset{(0.00895)}{0.0776}\ \texttt{smkban}},$$

$n = 10{,}000$, $R^2 = 0.0078$. Because `smkban` is the only regressor, the slope coefficient coincides with the raw difference in means from part (a): $\hat{\beta}_{\texttt{smkban}} = -0.0776$, and the intercept equals the cell mean for non-ban workers (0.2896). The robust t-statistic is $t = -0.0776/0.00895 = -8.67$ (equivalently, $F(1, 9998) = 75.06$, $p < 0.0001$), which rejects $H_0 : \beta_{\texttt{smkban}} = 0$ at any conventional significance level. The 95% CI is $[-0.095, -0.060]$.

The unconditional 7.8-pp gap should not be read as the causal effect of a ban. Workplaces that adopted bans in the early 1990s were disproportionately in office sectors with higher-education workforces, where smoking was already less prevalent, and part (c) repeats the exercise with controls to quantify how much of the raw gap is attributable to such composition effects. ■

(c) [8 points] Estimate a linear probability model with *smoker* as the dependent variable

and the following regressors: *smkban*, *female*, *age*, age^2, *hsdrop*, *hsgrad*, *colsome*, *colgrad*, *black*, and *hispanic*. Compare the estimated effect of a smoking ban from this regression with your answer from (b). Suggest an explanation, based on the substance of this regression, for the change in the estimated effect of a smoking ban between (b) and (c).

SOLUTION:

Table 8 reports both the simple LPM from (b) and the full-controls LPM. Holding demographic and education characteristics fixed, the coefficient on `smkban` falls (in absolute value) from -0.0776 to

$$\hat{\beta}^{\text{full}}_{\texttt{smkban}} = -0.0472 \qquad (\text{SE} = 0.0090),$$

a reduction of approximately 40%. The included controls absorb a large share of the raw gap through two substantive channels. Education comes first: the education coefficients are large and monotone, with smoking probability rising by 4.5 pp for college grads, 16.4 pp for some-college, 23.3 pp for high-school grads, and 32.3 pp for high-school dropouts relative to advanced-degree holders (all $p < 0.01$). Because the sample correlation between `smkban` and education is positive—bans were most common in high-education workplaces—failing to condition on education in part (b) biased $\hat{\beta}_{\texttt{smkban}}$ downward (that is, more negative). Demographics do similar work: smoking probability is 3.3 pp lower for women, 2.8 pp lower for Black workers, and 10.5 pp lower for Hispanic workers, and is modestly hump-shaped in age. These demographics are themselves correlated with ban coverage, contributing further to the attenuation.

Putting these two pieces together, of the raw 7.8-pp gap in part (b), roughly

3.0 pp reflects composition (selection into ban-covered workplaces), leaving a conditional ban effect of 4.7 pp. Under a conditional-independence assumption—the ban is "as good as random" given observed covariates—this -4.7 pp estimate is a credible causal effect. ■

(d) [5 points] Test the hypothesis that the coefficient on *smkban* is 0 in the population version of the regression in (c) against the alternative that it is nonzero, at the 5% significance level.

SOLUTION:

The test statistic is the robust F-statistic $F(1, 9989) = 27.76$ with $p < 0.0001$ (equivalently, $t = -\sqrt{27.76} = -5.27$). Because $27.76 \gg F_{0.05}(1, \infty) = 3.84$, we reject $H_0 : \beta_{\texttt{smkban}} = 0$ at the 5% level (and at any conventional level). The estimated ban effect of -4.7 pp is statistically distinguishable from zero with very high confidence. ■

(e) [6 points] Test the hypothesis that the probability of smoking does not depend on the level of education in the regression in (c). Does the probability of smoking increase or decrease with the level of education?

SOLUTION:

The null hypothesis is $H_0 : \beta_{\texttt{hsdrop}} = \beta_{\texttt{hsgrad}} = \beta_{\texttt{colsome}} = \beta_{\texttt{colgrad}} = 0$. The robust F-statistic is

$$F(4, 9989) = 140.09, \qquad p < 0.0001,$$

so we decisively reject. Education matters. Examining the individual coefficients,

$$\hat{\beta}_{\texttt{hsdrop}} = 0.323 \;>\; \hat{\beta}_{\texttt{hsgrad}} = 0.233 \;> \hat{\beta}_{\texttt{colsome}} = 0.164$$
$$> \;\hat{\beta}_{\texttt{colgrad}} = 0.0448 \;>\; 0 \;(\texttt{advanced, omitted}),$$

with all four coefficients significant at the 1% level. The probability of smoking *decreases monotonically with the level of education*: a high-school dropout is 28 pp more likely to smoke than a college graduate and 32 pp more likely than an advanced-degree holder, holding all other covariates fixed. Figure 4 visualizes this gradient. ■

Reflections: Although the point of SW 11.2 is to illustrate hypothesis testing in binary-response models, the substantive conclusion—strong education gradient in smoking—echoes the wider public-health literature. Evans et al. (1999) document similar patterns and estimate comparable workplace-ban effects (3–5 pp) using alternative empirical designs. The R^2 of the full model (0.057) is modest: education and demographics explain only a small fraction of who smokes, reflecting the importance of unmodeled factors like family history, peer effects, and local tobacco prices.

(f) [9 points] Repeat (c)–(e) using a probit model.

Solution:

Table 9 presents the probit estimates alongside the LPM. All coefficients retain their signs and their relative magnitudes follow the LPM ordering, but the probit coefficients are on the latent-index (z-score) scale and not directly interpretable

as probability changes. On the ban-effect magnitude from (c), the probit index coefficient on `smkban` is -0.159 (robust SE 0.0291); converting to an average partial effect via `margins, dydx(smkban)` gives

$$\overline{\partial \widehat{\Pr(\text{smoker})}/\partial \texttt{smkban}} = -0.0468 \text{ (SE} = 0.0086),$$

essentially identical to the LPM estimate of -0.0472. Because the probability of smoking in this sample sits in the 0.10–0.43 range—well inside the interior where probit and LPM agree (Angrist and Pischke, 2009, Ch. 3)—the two models produce indistinguishable average partial effects. The significance verdicts mirror the LPM's: the Wald $\chi^2(1) = 29.70$ ($p < 0.0001$) rejects $H_0 : \beta_{\texttt{smkban}} = 0$ at any conventional level, and the joint Wald $\chi^2(4) = 447.34$ ($p < 0.0001$) rejects the null that the four education dummies are jointly zero, with the monotone gradient in education identical to the LPM's.

The probit therefore does not overturn a single substantive conclusion from (c)–(e): the ban reduces smoking by roughly 5 pp, education is the strongest predictor, and smoking falls monotonically with education. This is the pattern Amemiya (1981) predicts—LPM and nonlinear latent-variable models typically deliver very similar average partial effects in the interior of the probability distribution—and it is the reason many applied economists (e.g. Angrist and Pischke, 2009) default to the LPM's computational simplicity and robust-SE interpretation unless the outcome is very rare or very common. ■

Appendix: Tables and Figures

Tables

Table 1: Determinants of Being an AmEx Cardholder (Problem 1 Table)

	(1) LPM	(2) Probit	(3) Logit	(4) LPM+drg	(5) Probit+drg	(6) Logit+drg
main						
Income ($1,000s)	0.00390***	0.0159***	0.0298***	0.00393***	0.0222***	0.0452***
	(0.000246)	(0.00121)	(0.00231)	(0.000227)	(0.00161)	(0.00316)
Age (years)	-0.00130***	-0.00441***	-0.00782***	-0.000236	0.000698	0.00107
	(0.000435)	(0.00146)	(0.00249)	(0.000395)	(0.00171)	(0.00311)
Self-employed = 1	-0.102***	-0.348***	-0.585***	-0.100***	-0.445***	-0.770***
	(0.0170)	(0.0536)	(0.0908)	(0.0157)	(0.0598)	(0.105)
Own residence = 1	0.0501***	0.174***	0.289***	0.0456***	0.191***	0.319***
	(0.00827)	(0.0291)	(0.0510)	(0.00741)	(0.0322)	(0.0586)
Months at current address	0.0000211	0.0000575	0.000139	0.000168***	0.00104***	0.00198***
	(0.0000613)	(0.000219)	(0.000377)	(0.0000543)	(0.000263)	(0.000476)
Major derogatory events (12 mo)				-0.117***	-0.706***	-1.234***
				(0.00431)	(0.0211)	(0.0374)
Minor derogatory events (12 mo)				-0.0317***	-0.0356*	-0.0695*
				(0.00517)	(0.0198)	(0.0360)
Constant	0.691***	0.416***	0.603***	0.712***	0.361***	0.420***
	(0.0134)	(0.0485)	(0.0852)	(0.0123)	(0.0580)	(0.107)
Observations	12,604	12,604	12,604	12,604	12,604	12,604
R^2	0.0269			0.2118		
Adjusted R^2	0.0265			0.2114		
Pseudo R^2		0.0284	0.0290		0.2296	0.2318
Log-likelihood	-6545.40	-6408.24	-6404.02	-5217.12	-5080.81	-5066.37

Standard errors in parentheses

Heteroskedasticity-robust standard errors in parentheses.

Stars: *p¡0.10, **p¡0.05, ***p¡0.01.

Dependent variable: cardhldr = 1 if applicant approved as cardholder.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Marginal Effects on Pr(cardhldr = 1) of `selfempl` and `ownrent`, by Income Level (reduced model)

	Absolute change $\Delta\hat{P}$			Percent change $\%\Delta\hat{P}$		
Income level	LPM	Probit	Logit	LPM	Probit	Logit
Δ*`selfempl` (switch from 0 to 1; `ownrent`=0, rest at medians)*						
20% income ($19.00k)	-0.1020	-0.1274	-0.1317	-0.1407	-0.1771	-0.1837
50% income ($26.16k)	-0.1020	-0.1209	-0.1222	-0.1355	-0.1599	-0.1613
80% income ($38.80k)	-0.1020	-0.1068	-0.1025	-0.1272	-0.1311	-0.1249
Δ*`ownrent` (switch from 0 to 1; `selfempl`=0, rest at medians)*						
20% income ($19.00k)	0.0501	0.0555	0.0549	0.0691	0.0771	0.0767
50% income ($26.16k)	0.0501	0.0511	0.0491	0.0666	0.0675	0.0648
80% income ($38.80k)	0.0501	0.0428	0.0388	0.0625	0.0525	0.0473

Notes. Other continuous covariates (`age`, `acadmos`) are held at their sample medians. Marginal effects are discrete differences in predicted probability from switching the indicator from 0 to 1. Percent changes are $\Delta\hat{P}/\hat{P}_0$.

Table 3: Change in Log-Odds of Approval, $\Delta \log[P/(1-P)]$, Across Income Levels

Change	Income level	LPM	Probit	Logit
$\Delta \log \frac{P}{1-P}$	*for switching* `selfempl` *from 0 to 1*			
	20%	-0.4668	-0.5691	-0.5847
	median	-0.4908	-0.5771	-0.5847
	80%	-0.5513	-0.5959	-0.5847
$\Delta \log \frac{P}{1-P}$	*for switching* `ownrent` *from 0 to 1*			
	20%	0.2677	0.2944	0.2894
	median	0.2907	0.3006	0.2894
	80%	0.3521	0.3140	0.2894

Notes. For the logit model these values should equal the dummy's regression coefficient (invariant to income); small departures reflect rounding.

Table 4: Effect of Age on the Probability of Employment in April 2009

	(1) LPM	(2) Probit	(3) Logit
main			
Age	0.0283***	0.122***	0.225***
	(0.00328)	(0.0129)	(0.0235)
age squared	-0.000327***	-0.00141***	-0.00262***
	(0.0000388)	(0.000155)	(0.000285)
Constant	0.307***	-1.258***	-2.490***
	(0.0669)	(0.250)	(0.446)
Observations	5,412	5,412	5,412
R^2	0.0197		
Pseudo R^2		0.0232	0.0235
Log-likelihood	-1628.68	-1986.92	-1986.43

Standard errors in parentheses

Heteroskedasticity-robust standard errors in parentheses.

Dependent variable = 1 if employed in April 2009, conditional on employment in April 2008.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Determinants of Employment Retention, April 2008 to April 2009

	(1) LPM	(2) Probit	(3) Logit
main			
Age	0.0253***	0.108***	0.197***
	(0.00355)	(0.0147)	(0.0272)
age squared	-0.000293***	-0.00127***	-0.00231***
	(0.0000419)	(0.000176)	(0.000327)
female	-0.00664	-0.0349	-0.0530
	(0.0100)	(0.0502)	(0.0942)
=1 if race != white	-0.0186	-0.0814	-0.146
	(0.0138)	(0.0632)	(0.117)
married	-0.00191	-0.00954	-0.0314
	(0.0104)	(0.0532)	(0.101)
union	-0.0577***	-0.291***	-0.533***
	(0.0149)	(0.0650)	(0.119)
ne_states	0.0164	0.0917	0.154
	(0.0143)	(0.0689)	(0.128)
so_states	0.0151	0.0719	0.134
	(0.0135)	(0.0641)	(0.119)
ce_states	0.0406***	0.209***	0.395***
	(0.0134)	(0.0687)	(0.130)
educ_lths	-0.0882***	-0.373***	-0.637***
	(0.0268)	(0.118)	(0.221)
educ_hs	-0.0251	-0.136	-0.233
	(0.0164)	(0.0924)	(0.179)
educ_somecol	-0.00509	-0.0361	-0.0542
	(0.0167)	(0.0970)	(0.189)
educ_aa	0.00316	0.0122	0.0379
	(0.0179)	(0.109)	(0.213)
educ_bac	-0.0162	-0.0976	-0.179
	(0.0152)	(0.0918)	(0.178)
Weekly earnings, Apr. 2008 (per 100 USD)	0.00348***	0.0217***	0.0444***
	(0.000953)	(0.00600)	(0.0125)
Constant	0.356***	-1.047***	-2.063***
	(0.0716)	(0.288)	(0.524)
Observations	4,773	4,773	4,773
R^2	0.0355		
Pseudo R^2		0.0451	0.0450
Log-likelihood	-1402.86	-1716.01	-1716.18

Standard errors in parentheses
Heteroskedasticity-robust standard errors in parentheses.
Omitted reference groups: white, western region, advanced degree.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Determinants of Becoming Unemployed by April 2009

	(1) LPM	(2) Probit	(3) Logit
main			
Age	-0.00506**	-0.0438**	-0.0898**
	(0.00232)	(0.0195)	(0.0415)
age squared	0.0000601**	0.000537**	0.00110**
	(0.0000271)	(0.000234)	(0.000499)
female	-0.0202***	-0.209***	-0.452***
	(0.00662)	(0.0666)	(0.145)
=1 if race != white	0.00983	0.104	0.192
	(0.00934)	(0.0837)	(0.179)
married	-0.0151**	-0.159**	-0.327**
	(0.00726)	(0.0705)	(0.155)
union	0.00119	-0.00468	0.0160
	(0.00875)	(0.0916)	(0.200)
ne_states	-0.0182**	-0.175*	-0.401*
	(0.00922)	(0.0946)	(0.206)
so_states	-0.00974	-0.0796	-0.192
	(0.00904)	(0.0836)	(0.178)
ce_states	-0.0143	-0.130	-0.306
	(0.00909)	(0.0887)	(0.194)
educ_lths	0.0699***	0.627***	1.347***
	(0.0189)	(0.161)	(0.361)
educ_hs	0.0339***	0.397***	0.892***
	(0.0100)	(0.134)	(0.314)
educ_somecol	0.0112	0.186	0.443
	(0.0101)	(0.143)	(0.337)
educ_aa	0.00936	0.145	0.338
	(0.0105)	(0.158)	(0.370)
educ_bac	0.0124	0.193	0.440
	(0.00897)	(0.137)	(0.323)
Weekly earnings, Apr. 2008 (per 100 USD)	-0.000378	-0.00433	-0.00892
	(0.000645)	(0.00738)	(0.0170)
Constant	0.158***	-0.823**	-1.295
	(0.0457)	(0.381)	(0.803)
Observations	4,773	4,773	4,773
R^2	0.0154		
Pseudo R^2		0.0371	0.0369
Log-likelihood	619.68	-887.65	-887.83

Standard errors in parentheses
Heteroskedasticity-robust standard errors in parentheses.
Dependent variable = 1 if unemployed in April 2009; =0 otherwise (including out-of-labor-force).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Determinants of Employment Retention, April 2006 to April 2007 (Normal Times)

	(1) LPM	(2) Probit	(3) Logit
main			
Age	0.0198***	0.0908***	0.166***
	(0.00341)	(0.0158)	(0.0300)
age squared	-0.000230***	-0.00108***	-0.00196***
	(0.0000403)	(0.000191)	(0.000365)
female	-0.0335***	-0.196***	-0.366***
	(0.00916)	(0.0544)	(0.106)
=1 if race != white	-0.0154	-0.0707	-0.143
	(0.0133)	(0.0713)	(0.136)
married	0.0225**	0.139**	0.250**
	(0.00990)	(0.0567)	(0.111)
union	-0.0119	-0.103	-0.188
	(0.0132)	(0.0800)	(0.156)
ne_states	0.00125	0.0161	0.0291
	(0.0136)	(0.0786)	(0.152)
so_states	0.00620	0.0414	0.0675
	(0.0125)	(0.0719)	(0.139)
ce_states	0.0182	0.109	0.218
	(0.0125)	(0.0743)	(0.144)
educ_lths	-0.0507**	-0.208	-0.376
	(0.0239)	(0.131)	(0.254)
educ_hs	-0.00802	-0.0362	-0.0532
	(0.0153)	(0.108)	(0.218)
educ_somecol	-0.00104	0.000465	0.0312
	(0.0164)	(0.115)	(0.230)
educ_aa	0.0118	0.0735	0.161
	(0.0171)	(0.125)	(0.252)
educ_bac	0.00140	0.00412	0.0142
	(0.0143)	(0.110)	(0.223)
Weekly earnings, Apr. 2006 (per 100 USD)	0.00357***	0.0305***	0.0675***
	(0.000824)	(0.00742)	(0.0164)
Constant	0.481***	-0.661**	-1.404**
	(0.0686)	(0.309)	(0.580)
Observations	4,569	4,569	4,569
R^2	0.0359		
Pseudo R^2		0.0535	0.0542
Log-likelihood	-903.55	-1408.07	-1407.05

Standard errors in parentheses
Heteroskedasticity-robust standard errors in parentheses.
Sample: workers employed full-time in April 2006.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Linear Probability Model of Smoking, Smoking.dta (n=10,000)

	(1) LPM simple	(2) LPM full controls
=1 if there is a work area smoking bans	-0.0776***	-0.0472***
	(0.00895)	(0.00897)
=1 if female		-0.0333***
		(0.00857)
age in years		0.00967***
		(0.00190)
age squared		-0.000132***
		(0.0000219)
=1 if hs dropout		0.323***
		(0.0195)
=1 if hs grad		0.233***
		(0.0126)
=1 if some college		0.164***
		(0.0126)
=1 if college grad		0.0448***
		(0.0120)
=1 if black		-0.0276*
		(0.0161)
=1 if hispanic		-0.105***
		(0.0140)
Constant	0.290***	-0.0141
	(0.00726)	(0.0414)
Observations	10,000	10,000
R^2	0.0078	0.0570
Adjusted R^2	0.0077	0.0560

Standard errors in parentheses
Heteroskedasticity-robust standard errors in parentheses.
Omitted education category: advanced degree.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: LPM vs Probit: Determinants of Smoking

	(1) LPM	(2) Probit
main		
=1 if there is a work area smoking bans	-0.0472***	-0.159***
	(0.00897)	(0.0291)
=1 if female	-0.0333***	-0.112***
	(0.00857)	(0.0288)
age in years	0.00967***	0.0345***
	(0.00190)	(0.00688)
age squared	-0.000132***	-0.000468***
	(0.0000219)	(0.0000826)
=1 if hs dropout	0.323***	1.142***
	(0.0195)	(0.0730)
=1 if hs grad	0.233***	0.883***
	(0.0126)	(0.0604)
=1 if some college	0.164***	0.677***
	(0.0126)	(0.0614)
=1 if college grad	0.0448***	0.235***
	(0.0120)	(0.0654)
=1 if black	-0.0276*	-0.0843
	(0.0161)	(0.0535)
=1 if hispanic	-0.105***	-0.338***
	(0.0140)	(0.0494)
Constant	-0.0141	-1.735***
	(0.0414)	(0.152)
Observations	10,000	10,000
R^2	0.0570	
Pseudo R^2		0.0544
Log-likelihood	-5420.74	-5235.87

Standard errors in parentheses
Heteroskedasticity-robust standard errors in parentheses.
Probit coefficients are index coefficients, NOT marginal effects.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figures

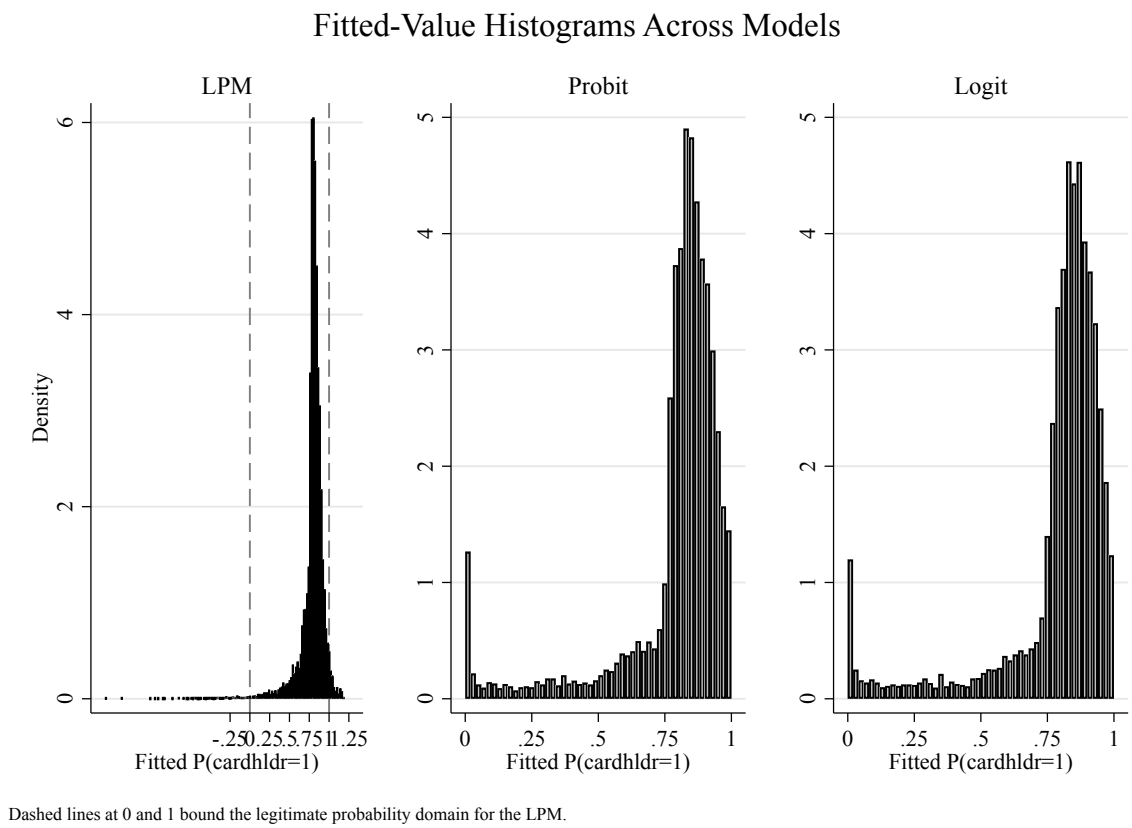

Figure 1: Histograms of fitted probabilities of being approved as a cardholder under the LPM, probit, and logit models with the full set of controls. The dashed lines in the LPM panel bound the legitimate probability domain $[0, 1]$; about 4.05% of LPM fitted values fall outside this range, while the probit and logit panels are bounded by construction. Source: own calculations using `AmEx.dta`.

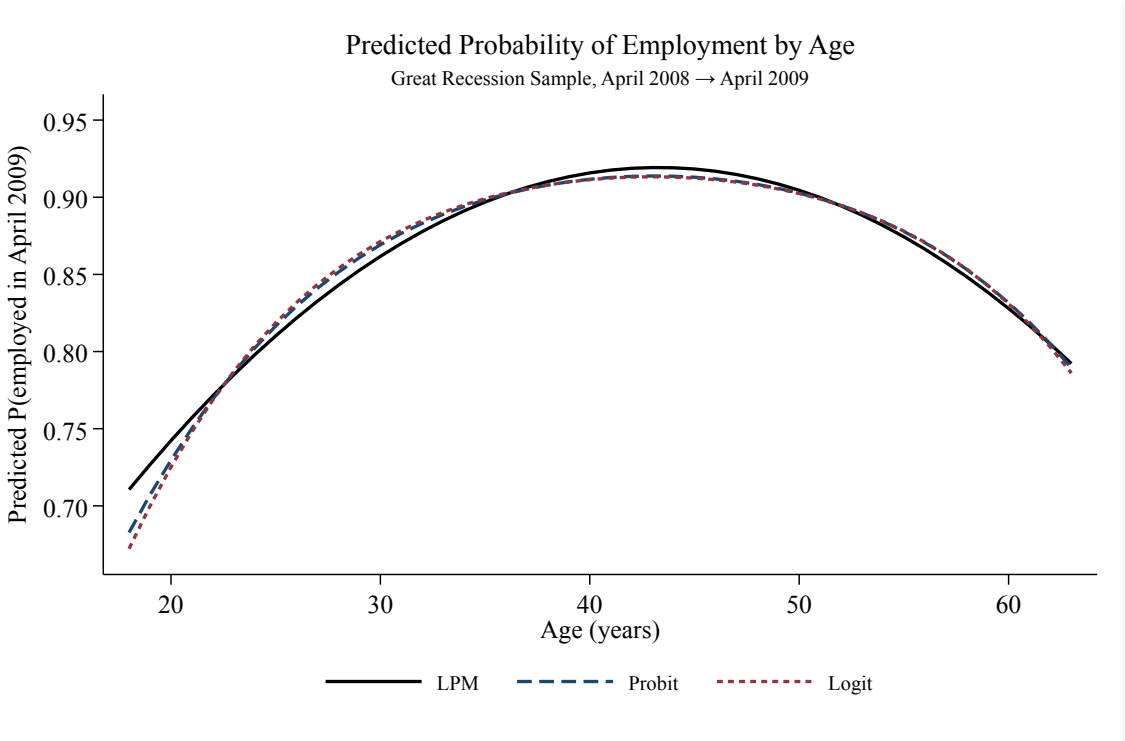

Figure 2: Predicted probability of remaining employed in April 2009, conditional on April 2008 full-time employment, as a function of age. LPM, probit, and logit predictions coincide almost everywhere inside the age support. Source: own calculations using `employment_08_09.dta`.

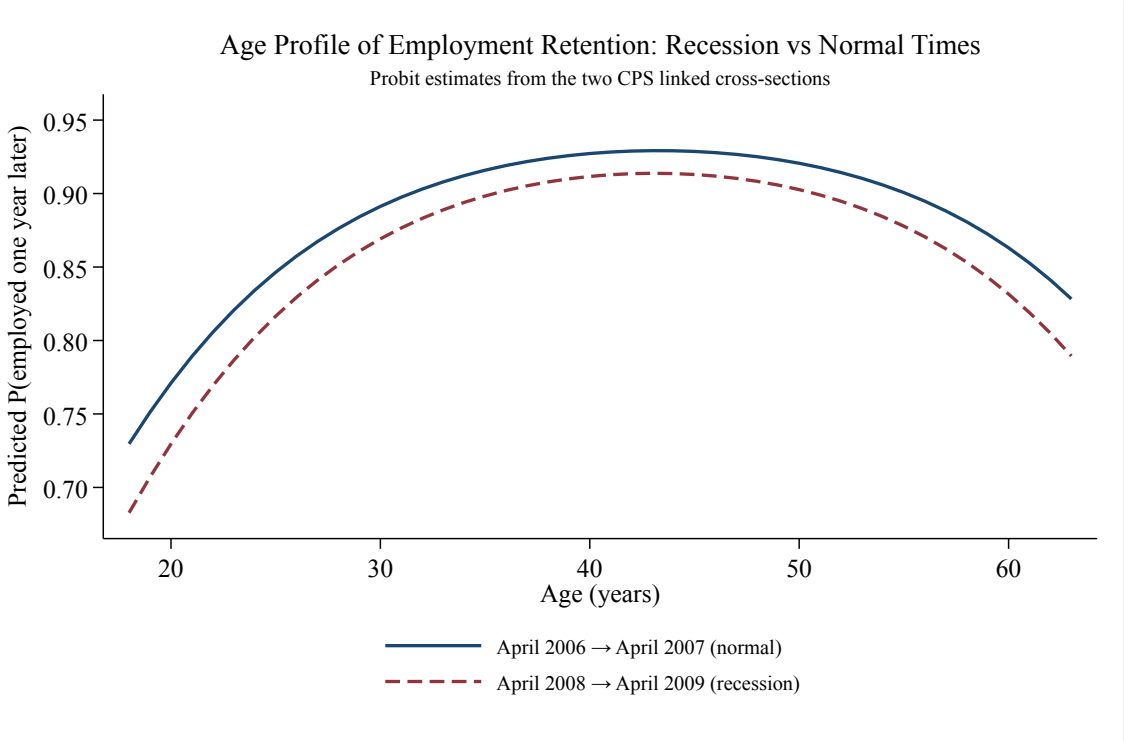

Figure 3: Probit-estimated age profile of employment retention for the Great Recession sample (2008–2009) compared to a normal-times benchmark (2006–2007). The recession lowers the entire curve by roughly 2 pp and slightly steepens it at the tails. Source: own calculations using `employment_06_07.dta` and `employment_08_09.dta`.

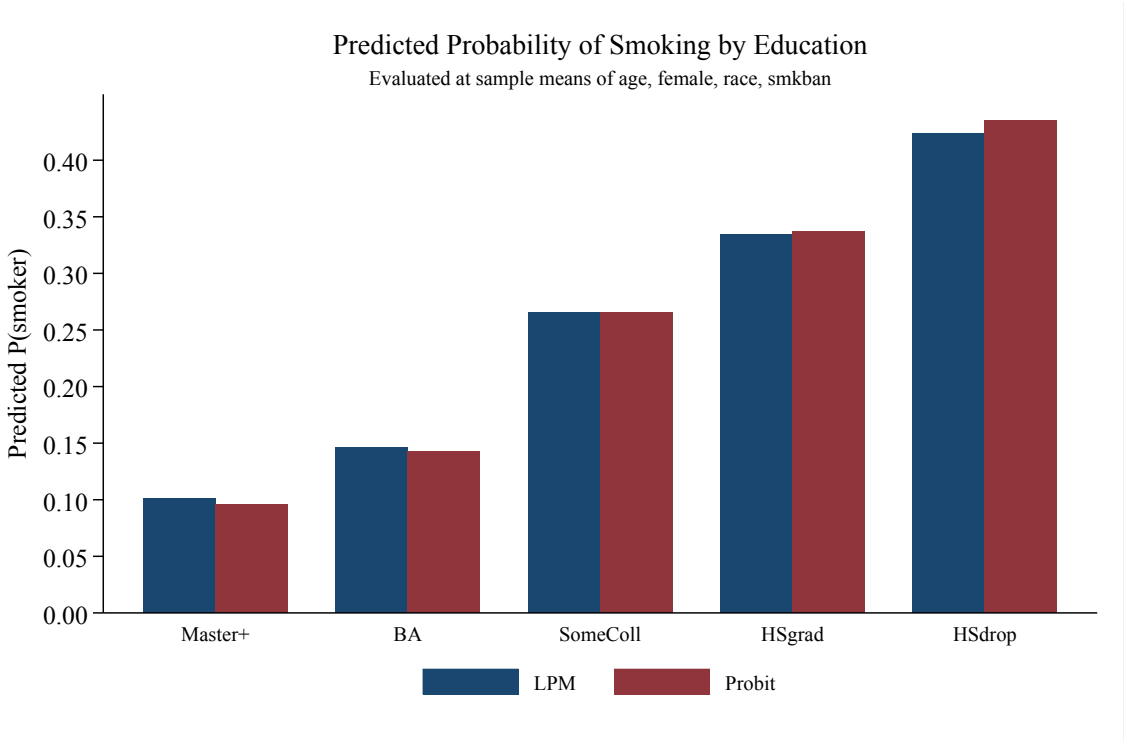

Figure 4: Predicted probability of smoking by education category, holding all other covariates at their sample means. LPM and probit yield indistinguishable predictions. The probability of smoking is monotonically decreasing in education. Source: own calculations using `smoking.dta`.

Replication Files

All replication materials for `homework_06` are available on GitHub:

- Course repository: https://github.com/tylersotomayor/econun3412-spring2026
- Homework 06 folder: https://github.com/tylersotomayor/econun3412-spring2026/tree/main/homework_06

The `do_files/` subfolder contains three heavily annotated Stata do-files: `ps6_amex.do` (Problem 1, parts a–n), `ps6_problem1.do` (SW 11.1), and `ps6_problem2.do` (SW 11.2). Each writes to a matching `.log` in the same directory. The `output/` subfolder contains the LaTeX tables (`output/tables/`) and PDF figures (`output/figures/`) generated by those do-files and `\inputed` into this document.

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